ORIGINDRIVE

Autonomous Navigation Project

Project Report & Operation Manual

1. Project Overview

Project Report: OriginDrive Robot Simulation

This document explains what each file in this project does in simple terms.

The ROS2 Package Setup Files ---
These files tell the computer how to build and organize the project so that it works as a standard ROS2 package.
* package.xml - Lists the name of the project ("origindrive") and the ROS2 tool libraries it needs to run (like rclpy and geometry_msgs).
* setup.py - The main instruction manual for installing the package. It tells ROS2 exactly where to look for the code and where the main starting point of the application is located.
* setup.cfg - A small configuration file that helps install the python scripts into the right, safe directory.

2. Core Component Analysis

2.1 Path Smoothing (path_smoother.py)

The Path Smoother takes a set of rough waypoints and turns them into a continuous curve. This is essential for robots because physical hardware cannot make instant 90-degree turns at speed.

Example: If a robot needs to go from (0,0) to (1,1) and then (2,0), a normal path is two straight lines. The smoother creates a rounded 'arc' at the corner, allowing the robot to maintain momentum.

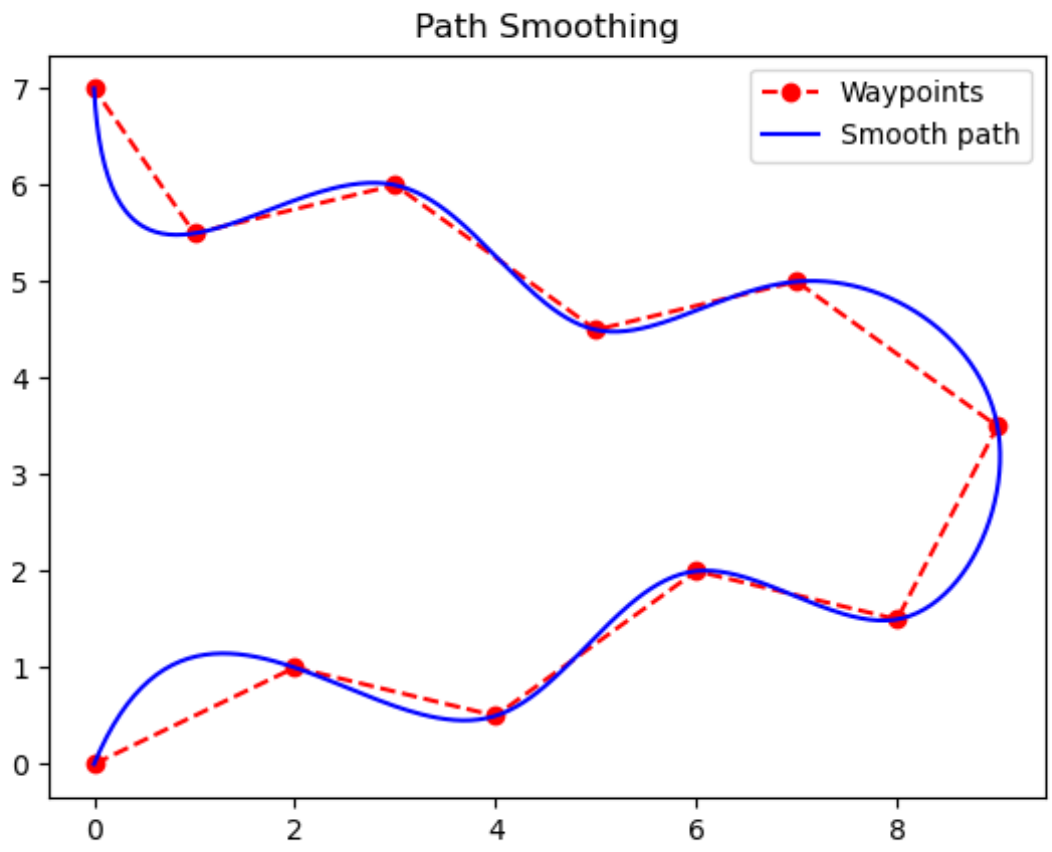

Figure 1: Smooth Path Generation vs Original Waypoints

2.2 Trajectory Planning (trajectory_generator.py)

This module calculates the speed profile of the robot. It ensures that the robot doesn't start or stop too abruptly, which could damage motors or tip the robot over.

Example: The robot smoothly ramps up speed to its 'V_MAX' and begins decelerating as it approaches the final goal, following a safe acceleration curve.

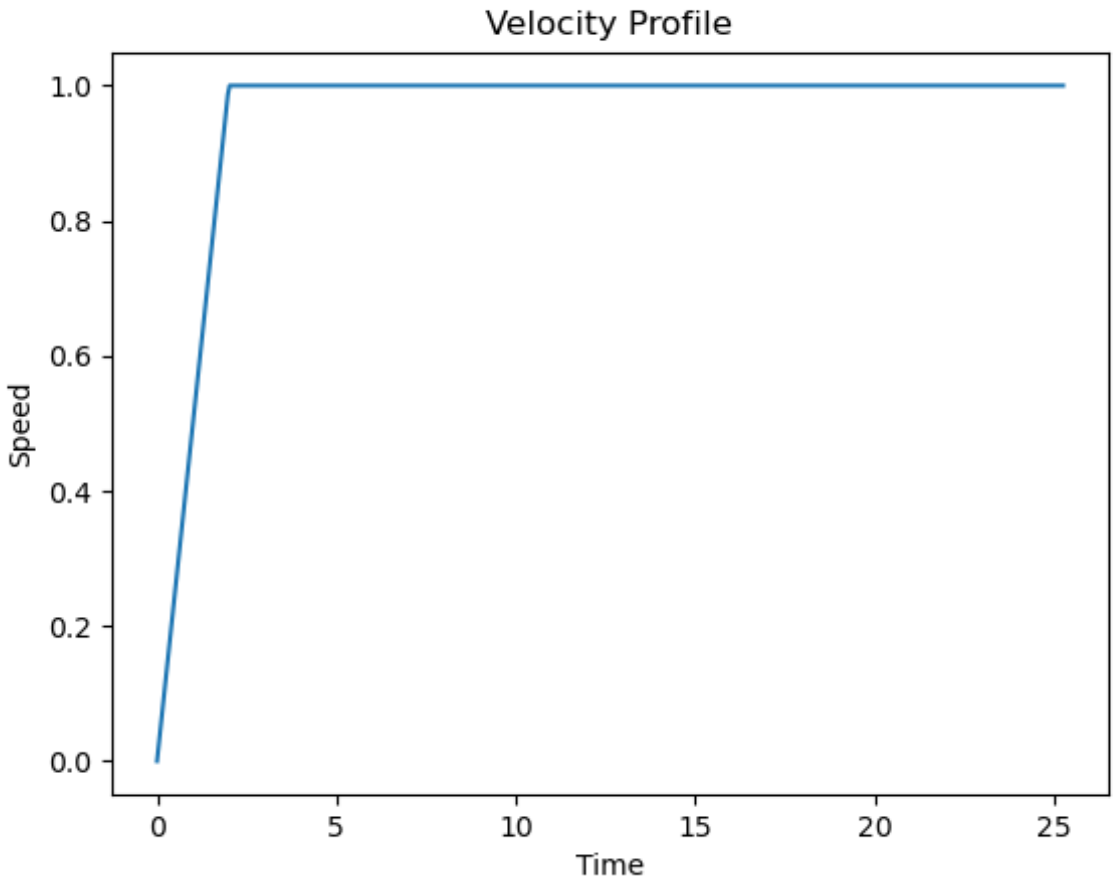

Figure 2: Planned Velocity Profile (Acceleration/Deceleration)

2.3 Pure Pursuit Control (controller.py)

The controller is the 'driver'. It uses the Pure Pursuit algorithm to look at a target point slightly ahead of the robot's current position and determines exactly how much to turn the wheels.

Example: As the robot follows the path, the controller calculates the 'curvature' needed to hit the next waypoint, adjusting for current heading and position in real-time.

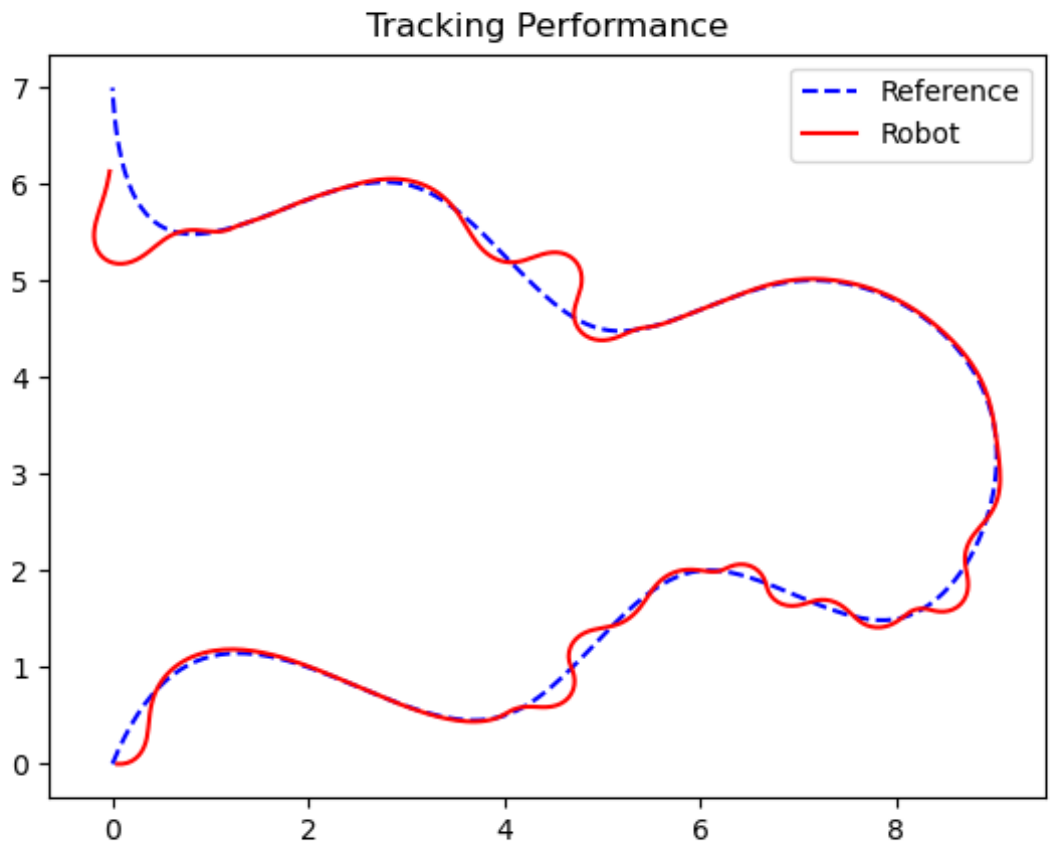

Figure 3: Simulated Robot Tracking Performance

2.4 Data Visualization (visualiser.py)

The Visualizer turns complex mathematical arrays into the clear, readable charts seen throughout this report. It helps engineers debug the navigation logic by showing exactly what happened during the drive.

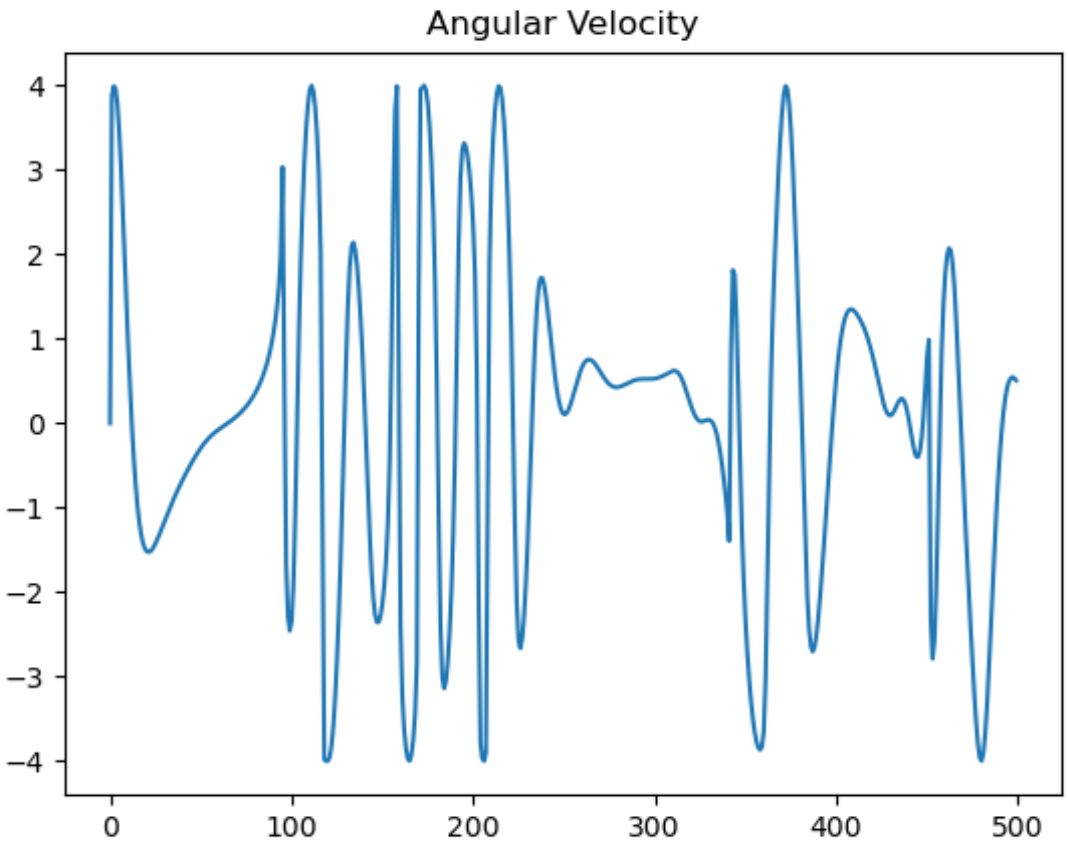

Figure 4: Angular Velocity Analysis

3. ROS2 Setup & Installation

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# ROS2 Humble Installation Commands for Ubuntu 22.04 (WSL)
# Open your Ubuntu/WSL terminal and run these commands step-by-step.

# Step 1: Setup Locale & Add ROS2 Source Repository
sudo apt update && sudo apt install locales
sudo locale-gen en_US en_US.UTF-8
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
export LANG=en_US.UTF-8

sudo apt install software-properties-common
sudo add-apt-repository universe

sudo apt update && sudo apt install curl -y
sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key -o /usr/share/keyrings/ros-archive-keyring.gpg

echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] http://packages.ros.org/ros2/ubuntu $(. /etc/os-release && echo $UBUNTU_CODENAME) main" | sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null

# Step 2: Install ROS2 & Colcon Build Tools
sudo apt update
sudo apt upgrade -y
sudo apt install ros-humble-desktop python3-colcon-common-extensions -y

# Step 3: Setup your bash environment
echo "source /opt/ros/humble/setup.bash" >> ~/.bashrc
source ~/.bashrc
```

4. Operations & Execution

1. Build the Package:
Run 'colcon build --packages-select origindrive' in your WSL terminal.

2. Source the Setup:
Load the environment with 'source install/setup.bash'.

3. Run the Node:
Launch the simulation with 'ros2 run origindrive navigation_node'.

The system will execute the autonomous navigation logic and generate the performance plots analyzed in Part 2 of this report.